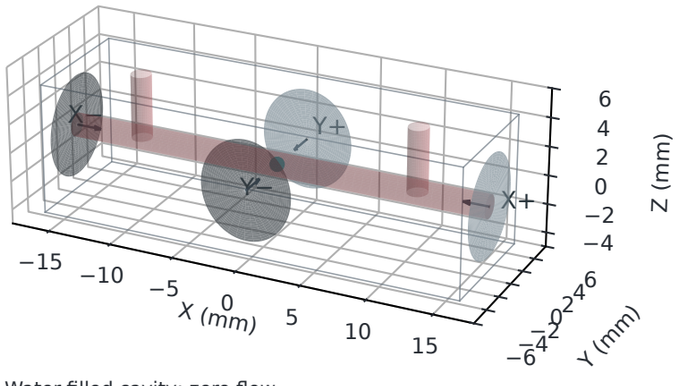
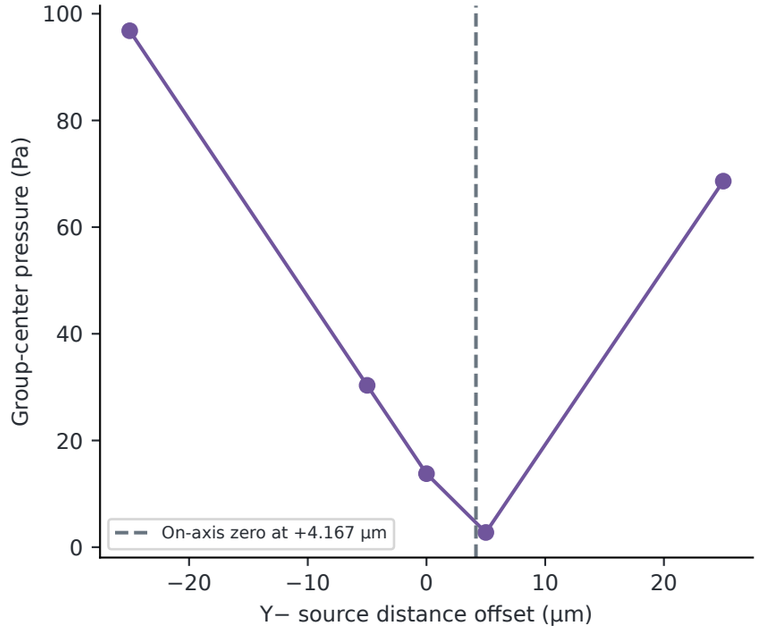
Y near-field sensitivity and PDMS field admission

a Confirmed apparatus geometry

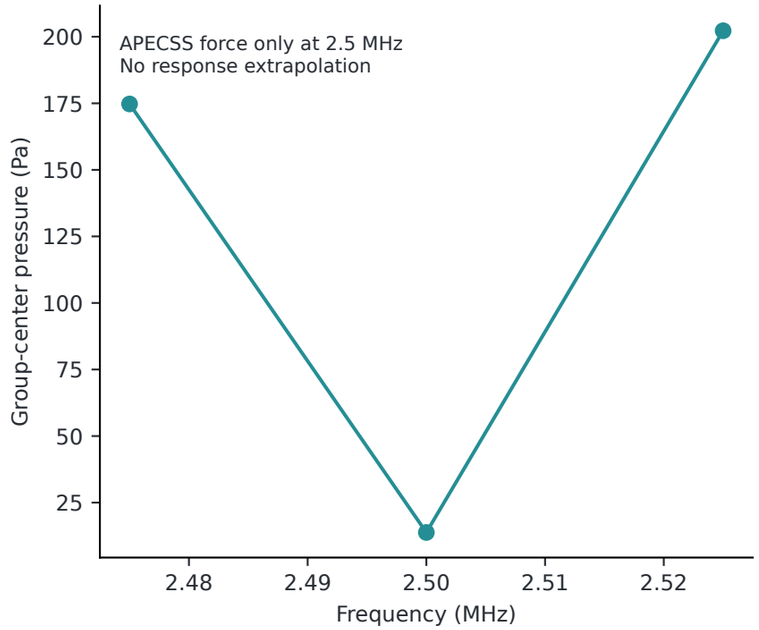


Water-filled cavity; zero flow
Both X ends sealed; 7 mm PZT discs

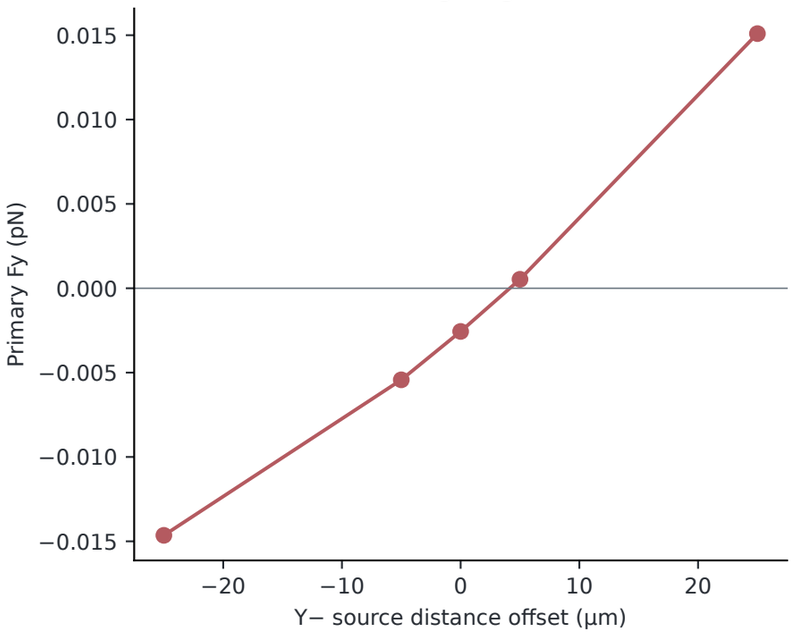
b Very near an axial pressure zero



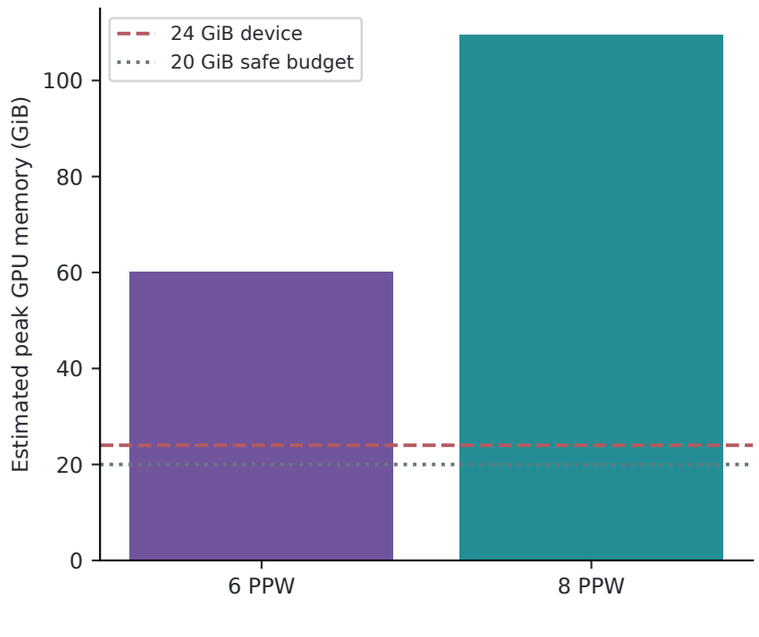
c Wavelength sensitivity (field only)



d Force can change sign near the zero



e Conditional 2.5 MHz resource estimate



f Material and resource admission

PDMS_FIELD_NOT_RUN
PDMS_Y_FIELD_NOT_ADMITTED

Xu et al., Phys. Rev. Applied 13, 054069 (2020)
Measured reference band: 3–7 MHz
2.5 MHz material input: not validated
Equivalent-fluid approximation omits shear waves

6 PPW: 68.1 µm, PML 20 cells, CFL ≤ 0.2
Grid incl. PML: 625 × 275 × 275
Estimated single-source runtime: 32.6 min
1 mm water envelope + BLI support margin
End-seal material / coupling uncalibrated

No multi-material pressure field is fabricated

All Y conclusions: HOMOGENEOUS_WATER_PROXY_SPECIFIC · Geometry offsets are exploratory, not measured assembly errors
EXPLORATORY · HOMOGENEOUS_WATER_CONTROL_PROXY · PHYSICS_UNCALIBRATED · training_ready=false